

Digital Logic Design

Lecture Topic: Boolean Algebra



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In the previous lecture

- Binary Logic

In the current lecture

- Boolean Algebra
- CLO 1: Understand number systems, codes, arithmetic and fundamentals of digital logic and Boolean algebra

Basic Definitions

- A **set** of elements is any collection of objects, usually having a common property.
- If S is a set, and x and y are certain objects,
 - $x \in S$ means that x is a member of the set S .
 - $y \notin S$ means that y is not an element of the set S .
- A **binary operator** defined on a set S of elements is a rule that assigns, to each pair of elements from S , a unique element from S .
 - $a*b=c$
 - $*$ is a binary operator if it specifies a rule for finding c from the pair (a,b) and also if $a, b, c \in S$.
 - $*$ is not a binary operator if $a, b \in S, c \notin S$.

Basic Definitions (cont.)

- Closure

- A set S is closed with respect to a binary operator if, for every pair of elements of S , the binary operators specifies a rule for obtaining a unique element of S .
 - The set of natural number $N=\{1,2,3,4,\dots\}$ **is** closed with respect to the binary operator $+$ by the rules of arithmetic addition, since, for any $a, b \in N$, there is a unique $c \in N$ such that $a+b=c$.
 - The set of natural number N **is not** closed with respect to the binary operator $-$ by the rules of arithmetic subtraction, since, $2-3=-1$, $2,3 \in N$, but $(-1) \notin N$.

Basic Definitions (cont.)

- Associative law

- A binary operator $*$ on a set S is said to be associative whenever

$$(x * y) * z = x * (y * z) \text{ for all } x, y, z \in S$$

- Commutative law

- A binary operator $*$ on a set S is said to be commutative whenever

$$x * y = y * x \text{ for all } x, y \in S$$

- Identity element

- A set S is said to have an identity element with respect to a binary operation $*$ on S if there exists an element $e \in S$ with the property that

$$e * x = x * e = x \text{ for every } x \in S$$

- The element 0 is an identity element with respect to binary operator $+$ on set of integers.

$$x + 0 = 0 + x = x$$

Basic Definitions (cont.)

- Inverse

- A set S having the identity element e with respect to a binary operator $*$ is said to have an inverse whenever, for every $x \in S$, there exists an element $y \in S$ such that

$$x * y = e$$

- In the set of integers, I , and the operator $+$, with $e=0$, the inverse of an element a is $(-a)$, since $a+(-a)=0$.

- Distributive law

- IF $*$ and $\cdot$ are two binary operators on a set S , $*$ is said to be distributive over $\cdot$ whenever

$$x * (y \cdot z) = (x * y) \cdot (x * z)$$

Axiomatic Definition of Boolean Algebra

- Boolean algebra
 - In 1854, George Boole developed an algebraic system.
- Switching algebra
 - In 1938, C. E. Shannon introduced a two-valued Boolean algebra called switching algebra that represented the properties of bistable electrical switching circuits.
- Huntington postulates
 - In 1904, E. V. Huntington introduced the postulates of Boolean algebra.
 - Boolean algebra is an algebraic structure defined by a set of elements, B , together with two binary operators, $+$ and $\cdot$.

Two-valued Boolean Algebra

- A two valued Boolean algebra is defined on a set of two elements, $B = \{0,1\}$, with rules for the operations $+$ and $\cdot$.

x	y	$x \cdot y$	x	y	$x+y$	x	x'
0	0	0	0	0	0	0	1
0	1	0	0	1	1	1	0
1	0	0	1	0	1		
1	1	1	1	1	1		

- The Huntington postulates are valid for the set B and the two binary operator $+$ and $\cdot$.
 - Closure
The result of each operation is either 1 or 0.
 - The identity elements
 - $+: 0$
 - $\cdot: 1$

Two-valued Boolean Algebra (cont.)

- The commutative laws are obvious from the symmetry of the binary operator tables.
- The distributive laws
 - $x(y+z) = xy + xz$ can be shown to hold from the operator tables.

x	y	z	$y + z$	$x \cdot (y + z)$	$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

Two-valued Boolean Algebra (cont.)

- **Complement**

- $x+x'=1$: since $0+0'=0+1=1$ and $1+1'=1+0=1$
- $x\cdot x'=0$: since $0\cdot 0'=0\cdot 1=0$ and $1\cdot 1'=1\cdot 0=0$

- Postulate 6 is satisfied because the two-valued Boolean Algebra has two distinct elements 1 and 0, with $0 \neq 1$

- **Note**

- a set of two elements
- $+$: OR operation; $\cdot$: AND operation
- a complement operator: NOT operation
- Binary logic is a two-valued Boolean algebra

Basic Theorems and Properties

- **Duality**

- the binary operators are interchanged; AND $\Leftrightarrow$ OR
- the identity elements are interchanged; 1 $\Leftrightarrow$ 0

Table 2.1

Postulates and Theorems of Boolean Algebra

Postulate 2	(a) $x + 0 = x$	(b) $x \cdot 1 = x$
Postulate 5	(a) $x + x' = 1$	(b) $x \cdot x' = 0$
Theorem 1	(a) $x + x = x$	(b) $x \cdot x = x$
Theorem 2	(a) $x + 1 = 1$	(b) $x \cdot 0 = 0$
Theorem 3, involution	$(x')' = x$	
Postulate 3, commutative	(a) $x + y = y + x$	(b) $xy = yx$
Theorem 4, associative	(a) $x + (y + z) = (x + y) + z$	(b) $x(yz) = (xy)z$
Postulate 4, distributive	(a) $x(y + z) = xy + xz$	(b) $x + yz = (x + y)(x + z)$
Theorem 5, DeMorgan	(a) $(x + y)' = x'y'$	(b) $(xy)' = x' + y'$
Theorem 6, absorption	(a) $x + xy = x$	(b) $x(x + y) = x$

Basic Theorems and Properties

- Theorem 1(a): $x+x = x$

$$\begin{aligned}x+x &= (x+x). 1 \\&= (x+x) (x+x') \\&= x+xx' \\&= x+0 \\&= x\end{aligned}$$

by postulate:

2(b)
5(a)
4(b)
5(b)
2(a)

- Theorem 1(b): $x . x = x$

$$\begin{aligned}xx &= x x + 0 \\&= xx + xx' \\&= x (x + x') \\&= x . 1 \\&= x\end{aligned}$$

Basic Theorems and Properties

- **Theorem 2 (a)**

- $x + 1 = 1$

$$\begin{aligned}x + 1 &= 1 \cdot (x + 1) \\&= (x + x')(x + 1) \\&= x + x' \cdot 1 \\&= x + x' \\&= 1\end{aligned}$$

- **Theorem 2 (b)**

$x \cdot 0 = 0$ by duality

- **Theorem 3: $(x')' = x$**

- Postulate 5 defines the complement of x , $x + x' = 1$ and $x \cdot x' = 0$
- The complement of x' is x is also $(x')'$

Basic Theorems and Properties

- Theorem 6 (a)

- $x + xy = x \cdot 1 + xy$
 $= x(1 + y)$
 $= x \cdot 1$
 $= x$

- **Theorem 6 (b)**

- $x(x + y) = x$ by duality

- By means of truth table

x	y	xy	$x + xy$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

Basic Theorems and Properties

- DeMorgan's Theorems

- $(x+y)' = x' y'$
- $(x y)' = x' + y'$

x	y	$x+y$	$(x+y)'$	x'	y'	$x' y'$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

Reading:

Page No. 38-45, Chapter 2

Digital Design with an Introduction to Verilog HDL, 5th Edition by Morris Mano

Thanks!